INCH-POUND

MIL-DTL-46027N (MR)
w/AMENDMENT 1
21 April 2025
SUPERSEDING
MIL-DTL-46027N (MR)
29 February 2024

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM ALLOY, WELDABLE 5083, 5456, and 5059

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers AA5083, AA5456, & AA5059 wrought aluminum alloy armor plate for both welded and unwelded applications in nominal thicknesses from 0.250 to 3.000 inch, inclusive with additional greater thicknesses for AA5083 from 3.001 to 10.500 (see 6.2).

1.2 Weldability. All three classes of material covered by this specification have demonstrated to be weldable to itself and other weldable alloys (see 6.3).

1.3 Classification. The wrought aluminum armor should be of the following classes as specified (see 6.2).

1.3.1 Class I. Class I is wrought aluminum armor that conforms to the Aluminum Association designation for the 5083 aluminum alloy. The applicable gauge range for class I is 0.250 - 10.500 inches.

1.3.2 Class II. Class II is wrought aluminum armor that conforms to the Aluminum Association designation for the 5456 aluminum alloy. The applicable gauge range for class II is 0.250 - 3.000 inches.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1.3.3 Class III. Class III is wrought aluminum armor that conforms to the Aluminum Association designation for the 5059 aluminum alloy. The applicable gauge range for class III is 0.250 - 3.000 inches.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V50 Ballistic Test for Armor
MIL-STD-3057	-	Arc Welding of Armor Grade Aluminum

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

TEST OPERATING PROCEDURES

TOP 2-2-711	-	Proofing Projectile Impact Testing of Armor Weldments

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE ALUMINUM ASSOCIATION, INC.

Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

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(Copies of these documents are available online at https://www.aluminum.org/).

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products

(Copies of these documents are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products
ASTM E716	-	Standard Practices for Sampling and Sample Preparation of Aluminum and Aluminum Alloys for Determination of Chemical Composition by Spark Atomic Emission Spectrometry
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Spark Atomic Emission Spectrometry
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)
ASTM G38	-	Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products

(Copies of these documents are available online at https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required, and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this

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specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical compositions for classes I, II, and III alloys defined by the Aluminum Association Teal Sheets, "International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys" are listed in table I and shall be within those limits. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

TABLE I. Chemical composition, percent. 1/

ELEMENTS	SYMBOL	CLASS I 5083 ALLOY 2/	CLASS II 5456 ALLOY 2/	CLASS III 5059 ALLOY 2/
Silicon	Si	0.40	0.25	0.45
Iron	Fe	0.40	0.40	0.50
Copper	Cu	0.10	0.10	0.25
Manganese	Mn	0.40 - 1.0	0.50 - 1.0	0.6 - 1.2
Magnesium	Mg	4.0 - 4.9	4.7 - 5.5	5.0 - 6.0
Chromium	Cr	0.05 - 0.25	0.05 - 0.20	0.25
Zinc	Zn	0.25	0.25	0.40 - 0.9
Titanium	Ti	0.15	0.20	0.20
Zirconium	Zr	---	---	0.05 - 0.25
Other, max. Each	---	0.05	0.05	0.05
Other, max. Total 3/	---	0.15	0.15	0.15
Aluminum	Al	Remainder	Remainder	Remainder

1/ Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those "others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.3 Mechanical properties. Unless otherwise specified in the contract or purchase order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal (L) direction shall meet the minimum mechanical properties listed in table II. If mechanical property requirements differ from those contained in table II, or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

TABLE II. Minimum mechanical properties. 1/,2/

Thickness, inches	Tensile Strength, ksi			Yield Strength, 0.2% Offset, ksi			Elongation percent		
	5083	5456	5059	5083	5456	5059	5083	5456	5059
0.250 to 0.499, incl.	45.0	54.0	53.0	35.0	47.0	39.0	8	6	8
0.500 to 2.000, incl.	45.0	50.0	52.0	37.0	45.0	39.0	8	6	7
2.001 to 3.000, incl.	44.0	50.0	47.0	35.0	44.0	39.0	9	8	8
3.001 to 10.500, incl.	43.0	TBD	TBD	31.0	TBD	TBD	11	TBD	TBD

1/ The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch.
2/ Values are taken in the Longitudinal Direction.

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3.4 Weldability. When the vehicle requires weldments, the fabricators (OEMs and kit manufacturers, etc.) are required to demonstrate weldability in accordance with MIL-STD-3057, "Arc Welding of Armor Grade Aluminum" for all armor to armor and armor to non-armor applications. This may also require Proofing Projectile Impact Testing (PPIT) for armor to armor weldments in accordance with procedures in TOP 2-2-711, "Proofing Projectile Impact Testing of Armor Weldments" (revision A and above) depending on the weld classification in accordance with MIL-STD-3057. PPIT procedures and associated striking velocities for established welded class I and class III plate thicknesses are listed in TOP 2-2-711. For any newly introduced armor material, configuration, or unproven thickness being used for armor applications, the aluminum fabricators shall qualify a weld procedure in accordance with MIL-STD-3057, and this requires PPIT testing following TOP 2-2-711. Since ordered thicknesses of the alloys referenced by this specification may not be listed in TOP 2-2-711, it is strongly suggested that the contract or drawing for the vehicle specify weldability requirements as outlined in paragraph 6.3.

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. No ballistic limit shall be specified for plate thickness of 0.250 to 0.499 inch, inclusive. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.5.1 Resistance to spalling. Material shall be free of spalling for those impact velocities of the V50 ballistic limit. The size of the spall that shall be considered a spall failure shall be no more than two calibers of the impacting projectile unless otherwise specified in the contract or purchase order (see 6.2). Spall that arises as part of the V50 ballistic limit shall be noted on the test data sheet, and test plates considered a failure due to spall shall also be reported as a failure to the ballistic requirements of specification MIL-DTL-46027 (see A.4.2).

3.6 Stress corrosion resistance. Stress Corrosion Cracking resistance is an optional requirement under this specification. When stress corrosion testing is requested in the contract or purchase order (see 6.2) and unless otherwise specified in the contract or purchase order (see 6.2), plate 0.750 inch and over in thickness shall be resistant to stress corrosion cracking, with a minimum of 5 of the 9 specimens showing no evidence of cracking at the end of 96 hours when tested as specified in 4.7.5.

3.7 Thermal processing. After final strain hardening performed during plate fabrication, any thermal processing exceeding 200°F (93°C) excluding fabrication by welding in accordance with MIL-STD-3057 shall necessitate recertification for conformance to 3.3, 3.5 and 3.6.

3.7.1 Storage of plates prior to application. After final strain hardening performed during plate fabrication, any extended outdoor storage (including container storage, e.g. CONEXs) exceeding 3 years shall necessitate recertification for conformance to 3.3, 3.5, and 3.6 by the entity owning the plates. MIL-DTL-46027 plates with unknown chain of custody shall also be recertified. For plates procured and qualified prior to the date of publication, this storage requirement while not mandatory is recommended. For optimal long term storage precautions that may obviate this storage and subsequent conformance requirement (see 6.5).

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3.8 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or purchase order (see 6.2).

3.8.1 Tolerances. Unless otherwise specified in the contract or purchase order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2/H35.2M, except for thickness.
3.8.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in table III.

TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		SPECIFIED WIDTH (Inches)								
		OVER	0.00	39.37	59.06	78.74	98.43	118.11	137.80	157.48
		THRU	39.37	59.06	78.74	98.43	118.11	137.80	157.48	177.17
OVER	THRU	TOLERANCES - INCHES (PLUS and MINUS)								
0.250	0.315		0.012	0.014	0.015	0.018	0.022	0.027	0.035	0.043
0.315	0.394		0.015	0.017	0.020	0.023	0.027	0.033	0.041	0.051
0.394	0.630		0.023	0.023	0.027	0.032	0.035	0.043	0.053	0.065
0.630	0.984		0.031	0.031	0.037	0.043	0.047	0.058	0.070	0.085
0.984	1.575		0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.575	2.362		0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.362	3.150		0.075	0.075	0.085	0.100	0.105	0.125	---	---
3.150	3.937		0.100	0.100	0.115	0.125	0.130	0.160	---	---
3.937	6.299		0.130	0.130	0.145	0.165	---	---	---	---
6.300	8.000		0.160	0.160	0.160	0.165	---	---	---	---
8.001	10.500		0.190	0.190	0.190	0.190	---	---	---	---

3.9 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Impression stamping shall not be used unless permitted by the procuring activity (see 6.2). Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall not contain more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.9.1 Ballistic test plates. In addition to the markings in 3.9, each ballistic test plate shall be marked with the letters ‘PRE’ for First Article test plates and ‘ACC’ for Acceptance test plates. This marking shall be impression stamped on the horizontal edge face of the ballistic plate in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see A.5.2).

3.10 Ballistic test plate information. For each lot of aluminum alloy armor, a properly completed Aluminum Armor Test Data Form (see figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

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3.11 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use. Minor scratches, gouges, dents and similar damage (see 3.11.2) are acceptable if the thickness is not reduced below the minimum material tolerance thickness.

3.11.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

3.11.2 Imperfections. The depth of surface scratches, nicks, pits, or gouges shall not reduce the plate thickness below the allowable minimum thickness. Isolated individual pits over 0.015 inches deep but not over 0.030 inches deep and not within 6 inches of each other and do not violate the minimum allowable thickness, are acceptable.

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY ARMOR									
SPECIFICATION: MIL-DTL-46027			**REVISION: N**			**AMENDMENT: 1**			
FIRING RECORD:		**DATE:**		**TEST CONTRACT NO:**					
	PLATE MANUFACTURER / PRODUCER			**PRIME CONTRACTOR**					
Name:									
Address:									
POC:									
Phone No:									
Email:									
TEST ITEM IDENTIFICATION:				Ordered Thickness:					
Lot No. _____ **Plate No.** _____			**Alloy:** _____ **Temper:** _____			**Class:** _____ **Type:** _____			
PURPOSE: ___ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing									
SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample _____)									
CHEMISTRY ANALYSIS:	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Zr**
V	**Li**	**Ag**	**Ni**	**Mn+Cr**	**Zr+Ti**	**Other - Each**			**Other - Total**	**Al**
										Rem.
MECHANICAL PROPERTIES:									
UTS (ksi):			0.2% YS (ksi):			Elongation (%):			
WELDABILITY:	Thickness		Required Velocity		Actual Velocity		Pass/Fail		
BALLISTIC TEST RESULTS: **SEPARATELY HEAT-TREATED TEST PLATE : YES ____ NO ____**									
	Obl. (deg)	**Actual Thickness (in)**	**Required V_{50} (fps)**	**Actual V_{50} (fps)**	**Pass/ Fail**				
Test:									
Projectile:									
	NOTES:								
SPALL:	Spall Size:		Two Projectile Calibers:		Pass / Fail:				
DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT & SUPPLIER OFFICIALS									
LOT: [____ met] [____ failed to meet] the ballistic requirements of specification MIL-DTL-46027									
Government Representative:	Date:	Supplier Representative:	Date:						

FIGURE 1. Aluminum Armor Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of all plate(s) of the same alloy and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate(s) in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit. All lots shall be fully traceable back to the ingot source by the supplier.

4.3 First article. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use in production.

4.3.1 First article inspection. First article inspection, except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examinations of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two (2) plates, 12 inches by 36 inches of each thickness to be supplied on the contract, shall be submitted for ballistic testing in accordance with Appendix A. The orientation of these plates with respect to the rolling direction shall be at the option of the producer (see 6.2). For ordered plate thicknesses exceeding 4.000”, the ballistic test plates shall be symmetrically machined from both top and bottom plate rolling surfaces to attain 4.000” (i.e. centered around the T/2 plane) with a thickness tolerance of +/- 0.060".

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4.5.1.4 Stress corrosion tests. When required (see 6.2), nine (9) stress corrosion test specimens shall be removed from the same plate that has been selected for ballistic testing. The location and type of specimen shall be as specified in 4.7.5.

4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

4.5.2.2.1 Thermal stability. When verification of thermal stability is required, mechanical testing of thermally stressed specimens shall be conducted. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot and thermally prepared prior to testing as directed in 4.7.2.1. Only one tension test specimen shall be made from each sample. The samples shall meet the minimum acceptance requirements for mechanical properties in table 2 when tested as specified in 4.7.2.

TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.3 Ballistic testing. One plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.4.

4.5.2.4 Stress corrosion tests. When required (see 6.2), nine test specimens shall be prepared from one plate randomly selected from each lot for stress corrosion testing. The plate shall meet the requirements when tested as specified in 4.7.5.

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4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.8) and workmanship (see 3.11) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.8 in accordance with the sampling procedures approved by the procuring activity and as specified in the contract or purchase order (see 6.2). At a minimum, one plate per lot shall be randomly selected for dimensional inspection.

4.7 Test specimens.

4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E716, E1251, and E3061. In case of any dispute, analysis by method E3061 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal (L) direction. For material less than 0.500 inch in thickness, a standard rectangular tension test specimen shall be used. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.2.1 Thermal stability specimens. When demonstration of thermal stability is required (see 4.5.2.2.1), mechanical properties of artificially aged specimens shall be determined (see 6.2(d)), specimens will follow the preparation of 4.7.2. The tensile specimens shall be artificially aged at 212°F (100°C) for 24 hours, removed and passively air cooled (without forced air) to 77°F (25°C), and then the tension tests shall be performed in accordance with ASTM B557.

4.7.3 Weldability. The thickness of the weldments specified in the contract or drawing shall be used to determine the proofing projectile and striking velocity in accordance with MIL-STD-3057 and TOP 2-2-711.

4.7.4 Ballistic testing. The ordered thickness specified in the contract or purchase order shall be used to determine the test projectile in accordance with table V. Ballistic testing shall be in accordance with Appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with table V and Appendix A to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL (P) for that plate.

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TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees	CLASS I, II, III
0.500 - 0.749	Cal. .30 AP	30	Table A-I
0.750 - 1.000	Cal. .50 FSP	0	Table A-II
1.001 - 1.700 1/	20-mm FSP	0	Table A-III
1.001 - 1.700 1/	Cal. .30 AP M2	0	Table A-IV
1.701 - 2.000	Cal. .30 AP M2	0	Table A-IV
2.001 - 10.500	Cal. .50 AP M2	0	Table A-V

1/ Two (2) types of projectiles for the same thickness range.

4.7.4.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to Rock Island Arsenal - Joint Manufacturing & Technology Center, Simulation & Test Range Division, Building 25 Prospect Court, Rock Island, IL 61299-5000 or to an approved Government facility for ballistic testing for first article or production acceptance. If Rock Island is unavailable or contraindicated in the contract or purchase order then the ballistic test plates shall be forwarded to Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059.

4.7.4.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

4.7.5 Stress corrosion specimens. When stress corrosion testing is required (see 6.2), short transverse stress corrosion test specimens shall be prepared in accordance with ASTM G38 and tested in accordance with ASTM G47. Nine specimens shall be tested per sample per lot. The specimens shall be stressed to 30 ksi. A maximum delay of 3 hours between stressing and initiation of stress corrosion test is permitted.

4.8 Rejection and retest. Unless otherwise specified in the contract or order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, 4.8.2, and 4.8.3.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection, except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails, the lot shall be permanently rejected with no further testing permitted.

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4.8.2 Ballistic retests. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.8.3 Stress corrosion retest. Immediately upon notification of the failure of the stress corrosion specimens, the manufacturer may at their expense test thirteen additional specimens for retesting. A minimum of seven specimens out of the thirteen additional specimens shall show no cracks upon examination at the end of 96 hours when tested as specified in 4.7.5. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2, 4.7.4 and 4.7.5.

4.9 Reduced testing. At the discretion of the procuring activity and as specified in the contract or purchase order (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing that the manufacturer agrees in writing to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat vehicles to protect the occupants against small arms fire, fragments, and shrapnel. Alloys 5083, 5456 and 5059 will be considered for multiple applications that require weldability, ballistic and blast performance.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify classification of alloy, and ordered thickness (see 1.1 and 1.3).
(c) When first article is required (see 3.1 and 4.3).
(d) Special mechanical properties (to include thermal stability mechanical properties) and ballistic requirements if different (see 3.3).
(e) If stress corrosion testing is required and if the test conditions are different (see 3.6, 4.5.2.1, 4.5.2.4, and 4.7.5).

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(f) Dimension and tolerance requirements, if different (see 3.8 and 3.8.1).
(g) If markings are different and when impression stamping is permissible (see 3.9).
(h) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.11.1).
(i) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(j) The orientation of the ballistic plate with respect to rolling is different (see 4.5.1.3 and 4.5.2.3).
(k) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(l) If approval was requested and received for a different ballistic testing facility (see 4.7.4.1).
(m) Rejection and retest requirements, if different (see 4.8).
(n) If reduced testing is allowed (see 4.9).
(o) Packaging requirements (see 5.1).
(p) Striking velocities for those thicknesses not covered (see MIL-STD-3057 and TOP 2-2-711).

6.3 Weldability. The U.S. Army DEVCOM, Ground Vehicle Systems Center, DEVCOM ARL, and the Aberdeen Test Center have conducted testing and determined that class I and class III alloys can be welded and demonstrate good shock resistance when prepared and tested in accordance with MIL-STD-3057 and TOP 2-2-711. If these alloys in thicknesses required for manufacturing are not currently listed in TOP 2-2-711, the fabricator should follow the procedures of TOP 2-2-711 to generate the required striking velocities before the Proofing Projectile Impact Testing (PPIT) is performed. The fabricator performing welding on these alloys for armor applications is responsible for developing and qualifying welding procedures in accordance with MIL-STD-3057 and TOP 2-2-711. Additional tests have demonstrated that material covered by this specification is weldable to itself and other weldable 7000, 5000, and 6000 series alloys using a 5000 series filler metal. It is not weldable to 2000 series alloys by conventional welding techniques. Any deviations from these conditions should be demonstrated in accordance with MIL-STD-3057 prior to acceptance by the procuring activity.

6.4 Density. The density of the specified alloys listed in this specification are as follows:

AA5083 = 0.096 lb/in^3 (2.66 g/cc)
AA5456 = 0.096 lb/in^3 (2.66 g/cc)
AA5059 = 0.096 lb/in^3 (2.66 g/cc)

6.5 Long Term Storage. Al-Mg (5XXX) alloys are susceptible to degradation of mechanical properties and corrosion resistance after extended durations at elevated temperatures beyond 131°F (55°C) and additional preparations for long term storage are recommended. Due to surface heating produced by direct sunlight and/or cumulative trapped heat in poorly ventilated CONEX type shipping and storage containers, ventilated indoor warehouse storage is recommended for optimal storage of MIL-DTL-46027 qualified plates. Outdoor storage that provides overhead shading from direct sunlight is also acceptable. Horizontally stacked plates should be separated by nonmetallic spacers to facilitate easy access and manipulation by heavy equipment and to provide optimum airflow to dissipate heat and maintain lower temperatures. Any MIL-DTL-

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46027 qualified plates with full documentation of the recommended long term storage methods listed above are eligible to waive the recertification requirements in 3.7.1.

6.6 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.7 Subject term (key word) listing.

Ballistic testing
Caliber .30 AP M2
Caliber .50 AP M2
Caliber .50 FSP
M1114 HMMWV
Military vehicles
Stress corrosion
20-mm FSP

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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APPENDIX A

ARMOR PLATE, ALUMINUM ALLOY, WELDABLE 5083, 5456, & 5059

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, weldable, 5083, 5456, and 5059 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification, and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of this document are available online at https://quicksearch.dla.mil/).

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any impact which is not a complete penetration may be considered a partial penetration.

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APPENDIX A

A.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-V.

A.4.2 Resistance to spalling. Material shall be free of spalling for those impact velocities of the V50 ballistic limit. The size of the spall that shall be considered a spall failure shall be no more than two calibers of the impacting projectile unless otherwise specified in the contract or purchase order (see 6.2). Spall that arises as part of the V_{50} ballistic limit shall be noted on the test data sheet, and test plates considered a failure due to spall shall also be reported as a failure to the ballistic requirements of specification MIL-DTL-46027.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the

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smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at their own expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at their own expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in A.5.3.1.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.435	1249	0.575	1459	0.715	1643
0.440	1257	0.580	1466	0.720	1649
0.445	1265	0.585	1473	0.725	1655
0.450	1273	0.590	1480	0.730	1661
0.455	1281	0.595	1487	0.735	1668
0.460	1290	0.600	1494	0.740	1674
0.465	1298	0.605	1500	0.745	1680
0.470	1305	0.610	1507	0.749 2/	1685
0.475	1313	0.615	1514	0.750	1686
0.480	1321	0.620	1521	0.755	1692
0.485	1328	0.625	1527	0.760	1699
0.490	1336	0.630	1534	0.765	1705
0.495	1343	0.635	1541	0.770	1711
0.500 1/	1351	0.640	1547	0.775	1717
0.505	1358	0.645	1554	0.780	1723
0.510	1366	0.650	1560	0.785	1728
0.515	1373	0.655	1567	0.790	1734
0.520	1381	0.660	1573	0.795	1739
0.525	1388	0.665	1580	0.800	1745
0.530	1395	0.670	1586	0.805	1751
0.535	1402	0.675	1593	0.810	1757
0.540	1410	0.680	1599	0.815	1763
0.545	1417	0.685	1605	0.820	1768
0.550	1424	0.690	1612	0.825	1774
0.555	1431	0.695	1618	0.830	1780
0.560	1438	0.700	1624	0.835	1786
0.565	1445	0.705	1630	0.840	---
0.570	1452	0.710	1637	0.845	---

1/ Specification requirements begin for this ordered thickness.
2/ Specification requirements end for this ordered thickness (See table A-II).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.665	1535	0.815	1978	0.965	2549
0.670	1547	0.820	1995	0.970	2570
0.675	1560	0.825	2012	0.975	2592
0.680	1573	0.830	2029	0.980	2614
0.685	1586	0.835	2046	0.985	2636
0.690	1599	0.840	2064	0.990	2658
0.695	1612	0.845	2081	0.995	2680
0.700	1625	0.850	2099	1.000 [2/]	2703
0.705	1639	0.855	2117	1.005	2726
0.710	1653	0.860	2135	1.010	2749
0.715	1667	0.865	2153	1.015	2772
0.720	1681	0.870	2171	1.020	2795
0.725	1696	0.875	2190	1.025	2819
0.730	1711	0.880	2209	1.030	2842
0.735	1725	0.885	2227	1.035	2866
0.740	1740	0.890	2246	1.040	2890
0.745	1755	0.895	2265	1.045	2914
0.750 [1/]	1770	0.900	2284	1.050	2939
0.755	1785	0.905	2304	1.055	2963
0.760	1801	0.910	2323	1.060	2988
0.765	1816	0.915	2343	1.065	3012
0.770	1832	0.920	2363	1.070	3037
0.775	1848	0.925	2383	1.075	3062
0.780	1863	0.930	2403	1.080	3087
0.785	1879	0.935	2423	1.085	3113
0.790	1895	0.940	2444	1.090	3139
0.795	1912	0.945	2464	1.095	3165
0.800	1928	0.950	2485	1.100	3191
0.805	1945	0.955	2506	1.105	3217
0.810	1961	0.960	2527	1.110	---

[1/] Specification requirements begin for this ordered thickness.
[2/] Specification requirements end for this ordered thickness (See table A-III and table A-IV).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20 mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.895	1271	1.060	1513	1.230	1811
0.900	1277	1.065	1521	1.235	1821
0.905	1284	1.070	1529	1.240	1830
0.910	1291	1.075	1537	1.245	1840
0.915	1298	1.080	1546	1.250	1849
0.920	1305	1.085	1554	1.255	1859
0.925	1311	1.090	1562	1.260	1869
0.930	1318	1.095	1570	1.265	1879
0.935	1325	1.100	1579	1.270	1889
0.940	1332	1.105	1587	1.275	1899
0.945	1339	1.110	1596	1.280	1909
0.950	1346	1.115	1604	1.285	1919
0.955	1353	1.120	1613	1.290	1929
0.960	1361	1.125	1621	1.295	1939
0.965	1368	1.130	1630	1.300	1949
0.970	1375	1.135	1638	1.305	1959
0.975	1382	1.140	1647	1.310	1970
0.980	1389	1.145	1656	1.315	1980
0.985	1397	1.150	1664	1.320	1991
0.990	1404	1.155	1673	1.325	2001
0.995	1412	1.160	1682	1.330	2012
1.000	1419	1.165	1691	1.335	2022
1.001[1/]	1420	1.170	1700	1.340	2033
1.005	1427	1.175	1709	1.345	2043
1.010	1435	1.180	1718	1.350	2054
1.015	1443	1.185	1727	1.355	2065
1.020	1450	1.190	1736	1.360	2076
1.025	1458	1.195	1745	1.365	2087
1.030	1466	1.200	1755	1.370	2098
1.035	1474	1.205	1764	1.375	2109
1.040	1481	1.210	1773	1.380	2120
1.045	1489	1.215	1783	1.385	2131
1.050	1497	1.220	1792	1.390	2142
1.055	1505	1.225	1801	1.395	2153

[1/] Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20 mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.400	2165	1.540	2505	1.680	2897
1.405	2176	1.545	2518	1.685	2912
1.410	2187	1.550	2531	1.690	2928
1.415	2199	1.555	2544	1.695	2943
1.420	2210	1.560	2558	1.700 2/	2958
1.425	2222	1.565	2571	1.705	2973
1.430	2234	1.570	2585	1.710	2989
1.435	2245	1.575	2598	1.715	3004
1.440	2257	1.580	2612	1.720	3019
1.445	2269	1.585	2625	1.725	3035
1.450	2281	1.590	2639	1.730	3051
1.455	2293	1.595	2653	1.735	3067
1.460	2305	1.600	2666	1.740	3083
1.465	2317	1.605	2680	1.745	3100
1.470	2329	1.610	2694	1.750	3116
1.475	2341	1.615	2708	1.755	3132
1.480	2353	1.620	2722	1.760	3149
1.485	2366	1.625	2737	1.765	3165
1.490	2378	1.630	2751	1.770	3182
1.495	2390	1.635	2765	1.775	3199
1.500	2403	1.640	2780	1.780	3215
1.505	2415	1.645	2794	1.785	3232
1.510	2428	1.650	2809	1.790	3249
1.515	2441	1.655	2823	1.795	3266
1.520	2453	1.660	2838	1.800	3283
1.525	2466	1.665	2853	1.805	3301
1.530	2479	1.670	2867	1.810	3318
1.535	2492	1.675	2882	1.815	3335

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.895	1681	1.040	1866	1.190	2038
0.900	1687	1.045	1872	1.195	2044
0.905	1694	1.050	1878	1.200	2050
0.910	1701	1.055	1884	1.205	2055
0.915	1707	1.060	1890	1.210	2060
0.920	1714	1.065	1896	1.215	2066
0.925	1720	1.070	1902	1.220	2071
0.930	1727	1.075	1907	1.225	2077
0.935	1733	1.080	1913	1.230	2082
0.940	1740	1.085	1919	1.235	2088
0.945	1746	1.090	1925	1.240	2093
0.950	1753	1.095	1931	1.245	2098
0.955	1759	1.100	1937	1.250	2104
0.960	1766	1.105	1942	1.255	2109
0.965	1772	1.110	1948	1.260	2114
0.970	1778	1.115	1954	1.265	2120
0.975	1785	1.120	1960	1.270	2125
0.980	1792	1.125	1965	1.275	2130
0.985	1798	1.130	1971	1.280	2136
0.990	1804	1.135	1977	1.285	2141
0.995	1810	1.140	1983	1.290	2146
1.000	1816	1.145	1988	1.295	2151
1.001 1/	1817	1.150	1994	1.300	2157
1.005	1823	1.155	2000	1.305	2162
1.010	1829	1.160	2005	1.310	2167
1.015	1836	1.165	2011	1.315	2172
1.020	1842	1.170	2016	1.320	2177
1.025	1848	1.175	2022	1.325	2183
1.030	1854	1.180	2027	1.330	2188
1.035	1860	1.185	2033	1.335	2193

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.340	2198	1.490	2347	1.640	2487
1.345	2203	1.495	2352	1.645	2492
1.350	2208	1.500	2356	1.650	2496
1.355	2213	1.505	2361	1.655	2501
1.360	2218	1.510	2366	1.660	2505
1.365	2223	1.515	2371	1.665	2510
1.370	2229	1.520	2376	1.670	2514
1.375	2234	1.525	2380	1.675	2519
1.380	2239	1.530	2385	1.680	2523
1.385	2244	1.535	2390	1.685	2528
1.390	2249	1.540	2394	1.690	2532
1.395	2254	1.545	2399	1.695	2537
1.400	2259	1.550	2404	1.700	2541
1.405	2264	1.555	2409	1.705	2545
1.410	2269	1.560	2413	1.710	2550
1.415	2273	1.565	2418	1.715	2554
1.420	2278	1.570	2423	1.720	2559
1.425	2283	1.575	2427	1.725	2563
1.430	2288	1.580	2432	1.730	2568
1.435	2293	1.585	2437	1.735	2572
1.440	2298	1.590	2441	1.740	2576
1.445	2303	1.595	2446	1.745	2581
1.450	2308	1.600	2450	1.750	2585
1.455	2313	1.605	2455	1.755	2590
1.460	2318	1.610	2460	1.760	2594
1.465	2322	1.615	2464	1.765	2598
1.470	2327	1.620	2469	1.770	2603
1.475	2332	1.625	2473	1.775	2607
1.480	2337	1.630	2478	1.780	2611
1.485	2342	1.635	2482	1.785	2616

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.790	2620	1.900	2713	2.010	2804
1.795	2624	1.905	2717	2.015	2808
1.800	2629	1.910	2722	2.020	2812
1.805	2633	1.915	2726	2.025	2816
1.810	2637	1.920	2730	2.030	2820
1.815	2641	1.925	2734	2.035	2824
1.820	2646	1.930	2738	2.040	2828
1.825	2650	1.935	2742	2.045	2832
1.830	2654	1.940	2746	2.050	2836
1.835	2658	1.945	2751	2.055	2840
1.840	2663	1.950	2755	2.060	2843
1.845	2667	1.955	2759	2.065	2847
1.850	2671	1.960	2763	2.070	2851
1.855	2675	1.965	2767	2.075	2855
1.860	2680	1.970	2771	2.080	2859
1.865	2684	1.975	2775	2.085	2863
1.870	2688	1.980	2779	2.090	2867
1.875	2692	1.985	2783	2.095	2871
1.880	2697	1.990	2787	2.100	2875
1.885	2701	1.995	2791	2.105	2879
1.890	2705	2.000 2/	2796	2.110	2882
1.895	2709	2.005	2800	2.115	2886

2/ Specification requirements end for this ordered thickness (See table A-V).

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.885	1995	2.040	2102	2.200	2206
1.890	1999	2.045	2105	2.205	2209
1.895	2002	2.050	2109	2.210	2212
1.900	2006	2.055	2112	2.215	2216
1.905	2009	2.060	2115	2.220	2219
1.910	2013	2.065	2119	2.225	2222
1.915	2016	2.070	2122	2.230	2225
1.920	2020	2.075	2125	2.235	2228
1.925	2023	2.080	2128	2.240	2231
1.930	2027	2.085	2132	2.245	2234
1.935	2030	2.090	2135	2.250	2238
1.940	2034	2.095	2138	2.255	2241
1.945	2037	2.100	2142	2.260	2244
1.950	2041	2.105	2145	2.265	2247
1.955	2045	2.110	2148	2.270	2250
1.960	2048	2.115	2151	2.275	2253
1.965	2051	2.120	2155	2.280	2256
1.970	2055	2.125	2158	2.285	2259
1.975	2058	2.130	2161	2.290	2263
1.980	2062	2.135	2164	2.295	2266
1.985	2065	2.140	2168	2.300	2269
1.990	2068	2.145	2171	2.305	2272
1.995	2072	2.150	2174	2.310	2275
2.000	2075	2.155	2177	2.315	2278
2.001 1/	2075	2.160	2180	2.320	2281
2.005	2079	2.165	2184	2.325	2284
2.010	2082	2.170	2187	2.330	2287
2.015	2085	2.175	2190	2.335	2290
2.020	2089	2.180	2193	2.340	2293
2.025	2092	2.185	2196	2.345	2296
2.030	2095	2.190	2200	2.350	2299
2.035	2099	2.195	2203	2.355	2303

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
2.360	2306	2.525	2404	2.690	2499
2.365	2309	2.530	2407	2.695	2501
2.370	2312	2.535	2410	2.700	2504
2.375	2315	2.540	2413	2.705	2507
2.380	2318	2.545	2416	2.710	2510
2.385	2321	2.550	2419	2.715	2513
2.390	2324	2.555	2421	2.720	2516
2.395	2327	2.560	2424	2.725	2518
2.400	2330	2.565	2427	2.730	2521
2.405	2333	2.570	2430	2.735	2524
2.410	2336	2.575	2433	2.740	2527
2.415	2339	2.580	2436	2.745	2529
2.420	2342	2.585	2439	2.750	2532
2.425	2345	2.590	2442	2.755	2535
2.430	2348	2.595	2445	2.760	2538
2.435	2351	2.600	2447	2.765	2541
2.440	2354	2.605	2450	2.770	2543
2.445	2357	2.610	2453	2.775	2546
2.450	2360	2.615	2456	2.780	2549
2.455	2363	2.620	2459	2.785	2552
2.460	2366	2.625	2462	2.790	2554
2.465	2369	2.630	2465	2.795	2557
2.470	2372	2.635	2468	2.800	2560
2.475	2375	2.640	2470	2.805	2563
2.480	2378	2.645	2473	2.810	2565
2.485	2380	2.650	2476	2.815	2568
2.490	2383	2.655	2479	2.820	2571
2.495	2386	2.660	2482	2.825	2574
2.500	2389	2.665	2485	2.830	2576
2.505	2392	2.670	2487	2.835	2579
2.510	2395	2.675	2490	2.840	2582
2.515	2398	2.680	2493	2.845	2585
2.520	2401	2.685	2496	2.850	2587

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
2.855	2590	3.020	2677	3.185	2751
2.860	2593	3.025	2679	3.190	2753
2.865	2595	3.030	2681	3.195	2755
2.870	2598	3.035	2684	3.200	2757
2.875	2601	3.040	2686	3.205	2759
2.880	2604	3.045	2688	3.210	2762
2.885	2606	3.050	2690	3.215	2764
2.890	2609	3.055	2693	3.220	2766
2.895	2612	3.060	2695	3.225	2768
2.900	2614	3.065	2697	3.230	2770
2.905	2617	3.070	2699	3.235	2772
2.910	2620	3.075	2702	3.240	2775
2.915	2622	3.080	2704	3.245	2777
2.920	2625	3.085	2706	3.250	2779
2.925	2628	3.090	2708	3.255	2781
2.930	2630	3.095	2711	3.260	2783
2.935	2633	3.100	2713	3.265	2786
2.940	2636	3.105	2715	3.270	2788
2.945	2639	3.110	2717	3.275	2790
2.950	2641	3.115	2720	3.280	2792
2.955	2644	3.120	2722	3.285	2794
2.960	2647	3.125	2724	3.290	2796
2.965	2649	3.130	2726	3.295	2799
2.970	2652	3.135	2728	3.300	2801
2.975	2654	3.140	2731	3.305	2803
2.980	2657	3.145	2733	3.310	2805
2.985	2660	3.150	2735	3.315	2807
2.990	2662	3.155	2737	3.320	2809
2.995	2665	3.160	2740	3.325	2811
3.000 [2/]	2668	3.165	2742	3.330	2814
3.005	2670	3.170	2744	3.335	2816
3.010	2672	3.175	2746	3.340	2818
3.015	2675	3.180	2748	3.345	2820

[2/] Specification class II and class III specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
3.350	2822	3.495	2884	3.640	2944
3.355	2824	3.500	2886	3.645	2946
3.360	2827	3.505	2888	3.650	2948
3.365	2829	3.510	2890	3.655	2950
3.370	2831	3.515	2892	3.660	2952
3.375	2833	3.520	2894	3.665	2954
3.380	2835	3.525	2896	3.670	2956
3.385	2837	3.530	2898	3.675	2958
3.390	2839	3.535	2901	3.680	2960
3.395	2841	3.540	2903	3.685	2962
3.400	2844	3.545	2905	3.690	2964
3.405	2846	3.550	2907	3.695	2967
3.410	2848	3.555	2909	3.700	2969
3.415	2850	3.560	2911	3.705	2971
3.420	2852	3.565	2913	3.710	2973
3.425	2854	3.570	2915	3.715	2975
3.430	2856	3.575	2917	3.720	2977
3.435	2858	3.580	2919	3.725	2979
3.440	2861	3.585	2921	3.730	2981
3.445	2863	3.590	2923	3.735	2983
3.450	2865	3.595	2925	3.740	2985
3.455	2867	3.600	2928	3.745	2987
3.460	2869	3.605	2930	3.750	2989
3.465	2871	3.610	2932	3.755	2991
3.470	2873	3.615	2934	3.760	2993
3.475	2875	3.620	2936	3.765	2995
3.480	2877	3.625	2938	3.770	2997
3.485	2880	3.630	2940	3.775	2999
3.490	2882	3.635	2942	3.780	3001

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
3.785	3003	3.925	3059	4.065	3114
3.790	3005	3.930	3061	4.070	3116
3.795	3007	3.935	3063	4.075	3118
3.800	3009	3.940	3065	4.080	3120
3.805	3011	3.945	3067	4.085	3122
3.810	3013	3.950	3069	4.090	3124
3.815	3015	3.955	3071	4.095	3125
3.820	3017	3.960	3073	4.100	3127
3.825	3019	3.965	3075	4.105	3129
3.830	3021	3.970	3077	4.110	3131
3.835	3023	3.975	3079	4.115	3133
3.840	3025	3.980	3081	4.120	3135
3.845	3027	3.985	3083	4.125	3137
3.850	3029	3.990	3085	4.130	3139
3.855	3031	3.995	3087	4.135	3141
3.860	3033	4.000 3/,4/	3088	4.140	3143
3.865	3035	4.005	3090	4.145	3145
3.870	3037	4.010	3092	4.150	3147
3.875	3039	4.015	3094	4.155	3149
3.880	3041	4.020	3096	4.160	3151
3.885	3043	4.025	3098	4.165	3152
3.890	3045	4.030	3100	4.170	3154
3.895	3047	4.035	3102	4.175	3156
3.900	3049	4.040	3104	4.180	3158
3.905	3051	4.045	3106	4.185	3160
3.910	3053	4.050	3108	4.190	3162
3.915	3055	4.055	3110	---	---
3.920	3057	4.060	3112	---	---

3/ Specification class I specification requirements end for this ordered thickness.
4/ For plates ordered in 4.001" to 6.000", deduct 70 fps from the required BL(P) of the final machined thickness; for plates ordered in 6.001" to 10.500", deduct 100 fps from the required BL(P) of the final machined thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY - MR

(Project 9535-2025-007)

Review activities:
Army - AR, AT, AV, TE

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